Notation

R. Heller · Date: 2026-05-11 · Last changed: 2026-05-11

Symbol	Meaning
N	Set of players, $N = \{1, 2, \dots, n\}$
A_i	Action set of player i
$A = \prod_i A_i$	Joint action space
a_{-i}	Action profile of all players except i
$u_i : A \to \mathbb{R}$	Payoff (utility) function of player i
$\sigma_i \in \Delta(A_i)$	Mixed strategy of player i
$\$ \widehat{} * \$$	Nash equilibrium profile
$v(S)$	Characteristic function of coalition S in cooperative games
$\phi_i(v)$	Shapley value of player i
π	Policy in reinforcement learning
$Q(s, a)$	Action–value function
γ	Discount factor
$\mathbb{E}[\cdot]$	Expectation

Common abbreviations

Abbrev.	Expansion
BR	Best response
BNE	Bayesian Nash equilibrium
CFR	Counterfactual regret minimisation
ESS	Evolutionarily stable strategy
MARL	Multi-agent reinforcement learning
MDP	Markov decision process
MSNE	Mixed-strategy Nash equilibrium
NE	Nash equilibrium
PD	Prisoner's dilemma

Abbrev.	Expansion
RL	Reinforcement learning
SPE	Subgame-perfect equilibrium
TU / NTU	Transferable / non-transferable utility
VCG	Vickrey–Clarke–Groves mechanism